Ghofrane Mahfoudhi

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Summary

ICT Engineering Student – Specialized in Generative AI, Autonomous Agents, and Deep Learning
Final-year student at SUPCOM, passionate about developing **autonomous intelligent systems**, **generative models**, and **classical AI solutions** like **deep learning**. Seeking a 6-month internship to contribute to innovative AI projects, whether in **intelligent agents**, **generative AI**, or **deep learning applications**.Proven experience in designing and deploying AI agents with LangChain, as well as deep learning modeling for image detection and analysis.

Education

- **Higher School of Communications of Tunis, SUP'COM** — *2023 - Ongoing*
  ICT Engineering — Ariana, Tunisia
- **Preparatory Institute for Engineering Studies of Sfax** — *2021 - 2023*
  Preparatory Classes - Ranked 24/1500 in National Engineering Competition — Sfax, Tunisia

Professional Experience

- **KPIT Technologies** — *July 2025 – August 2025*
  AI / Data Engineering Intern — Sfax, Tunisia
  - **Automated Data Engineering Pipeline with LLMs & AI Agents :** Design and development of an intelligent system to automate the generation of SQL queries and Grafana dashboards from natural language descriptions. Implementation of a ReAct agent (LangChain) powered by the Mistral LLM, capable of iterative reasoning to produce complex and accurate SQL queries. Development of a dashboard generator using advanced prompt engineering to produce a valid and importable JSON file.
  - **Technologies:** Python, LangChain, Mistral AI API, SQL, Grafana API, Prompt Engineering, Agents IA
- **Numeryx MEA** — *July 2024 – August 2024*
  Development Intern — Sfax, Tunisia
  - **Full-stack development** of the "CAR PARK" **web app** for automated vehicle and client management. Built a **MySQL database**, responsive **Bootstrap interface**, and **PHP-based CRUD features** with secure user authentication.
  - **Technologies:** HTML, CSS, Bootstrap, PHP, MySQL, XAMPP
- **SUP'COM** — *May 2025*
  Entrepreneurship Project Challenge — Tunis, Tunisia
  - Selected among the **8 finalists** for a week-long challenge. **Complete project management**: market research, benchmarking, financial analysis, and presentation of a Business Model Canvas for a connected intelligent patch for real-time dehydration detection.

Key Projects

- **AI Assistant for Agricultural Risk Management (Ongoing)** — *September 2025 – Ongoing*
  - Developing an **intelligent AI agent** using LangChain and LLaMA-2 to **analyze dynamic environments** (weather data, risk factors) and **recommend optimized decisions** (insurance coverage) for farmers.
  - **Testing and evaluating agent behaviors** to ensure effective interaction with real-world data.
  - **Technologies:** Weather APIs, LangChain Agents, LLaMA-2, Gradio
- **AI Chatbot for Energy Optimization (Ongoing)** — *September 2025 – Ongoing*
  - Building an **AI chatbot** for smart grids using **LLMs** to **optimize domestic energy in real-time**, provide consumption data analysis and energy-saving advice.
- **AI-Powered Human Life Detection System for Search and Rescue Operations** — *Jan 2025 - June 2025*
  - Designed and Developed an **AI-powered survival detection** system using **FMCW radar** and **thermal imaging**. Processed **radar signals** with **CWT/STFT** and applied a **deep learning fusion model** (CNNs, 3D-CNNs, and an MLP for decision fusion) to achieve 99% accuracy in identifying survivors under debris, enhancing search and rescue efficiency.
  - **Technologies:** Deep Learning, CNN, MATLAB, Python, TensorFlow
- **EduBot – Intelligent Learning Assistant** — *Nov 2024 - Dec 2024*
  - Development of a **web platform** with intuitive interface for educational content, integrated with **AI chatbot** for course questions.
  - **Technologies:** Java Spring Boot, Angular, PostgreSQL, AI API

Skills

Programming: Python, C++, MATLAB
Cloud: Amazon Web Services AWS
Web: Java, PHP, Angular, HTML, CSS
Data & Visualization: Power BI, Grafana

Soft Skills: Teamwork, Fast Learning, Autonomy, Clear Communication
Interests: Volleyball, cinema, cooking, outdoor activities (zip-lining, jet-ski, karting)

Volunteering

- **PACTE PROJECT - Community project management for high school infrastructure improvement** — *2024*
- **Collaboration Radio Jeune Tunisie - Production of a radio report** — *2024*
- **IEEE SUP'COM - Member of RAS Committee (Robotics and Automation Society)** — *2023*
- **Forum Sup'Com 2023 - Co-Design Manager** — *2023*

Additional Information

Languages: Arabic (Native), English (B2), French (B2)